

Hybrid Search

Sparse Vectors

+

Dense Vectors

RRF
Rank Reciprocal Fusion

BM25 → Sparse Vectors

OPENAI → Dense Vectors

Combining

Weighted sum of normalized scores

$$\alpha * \text{vector scores} + 1 - \alpha * \text{bm25 scores}$$

Normalization

S Min Max Normalization BM25 scores
D Invert & normalize vector scores

Normalize scores

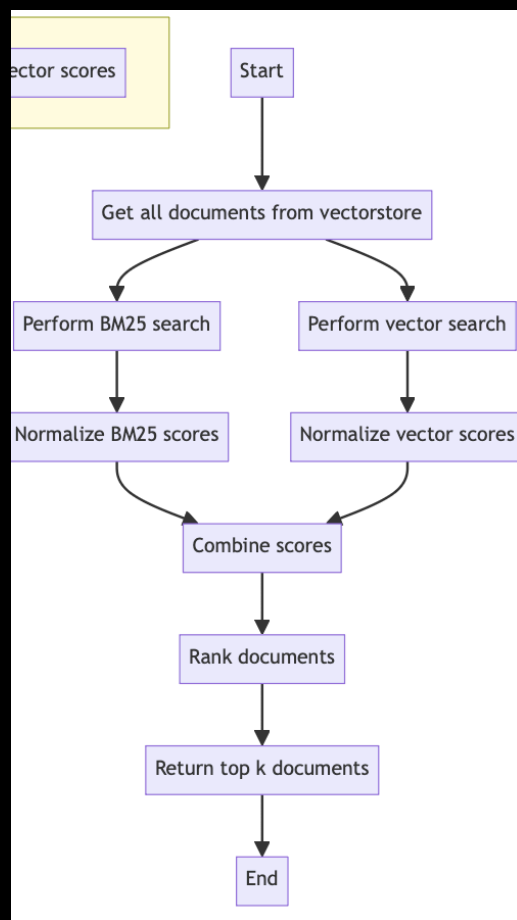
Min-max normalization for BM25

Invert and normalize vector scores

Combine scores

Weighted sum of normalized scores

$\alpha * \text{vector_scores} + 1 - \alpha * \text{bm25_scores}$



Focus:- Better Quality Sparse Vectors

TF-IDF, BM25

- 1) Document length
- 2) Keyword Saturation

Static → Formula
↘
learnable

SPLADE

BERT

MLM

Learnable Sparse Vector Masked Language Model

Tokenatization

Token

Fill in the blanks

Query $\rightarrow$ Sub word level Tokenization

Predict the most preferable token.

MLM

BERT Vocab Size:- 30,552 tokens

Context of the entire sequence.

Splade will combine

1) Probabilities :- across each token

Encoder

prefer

$\downarrow$

probable

Logarithmic Saturation

Logic 0
Constant

↓
Data sparsity

Dim:- 30,572

1000, 30000

Learnable Sparse Vector
Static ↗